1. Which structural form of DNA represents the standard Watson-Crick double helix under physiological conditions?

(A) B-DNA
(B) A-DNA
(C) Z-DNA
(D) C-DNA

2. In Watson-Crick B-DNA, how many base pairs are present per complete helical turn?

(A) 10.0 to 10.5 bp
(B) 8.0 bp
(C) 12.0 bp
(D) 14.5 bp

3. Which nitrogenous bases belong to the purine family?

(A) Adenine and Guanine
(B) Cytosine and Thymine
(C) Uracil and Cytosine
(D) Thymine and Guanine

4. According to Chargaff's rules, in double-stranded DNA the molar ratio of A+G equals:

(A) T+C (Purines = Pyrimidines)
(B) T+U
(C) G+C
(D) 2*(A+T)

5. Z-DNA is unique among DNA conformations because it forms a:

(A) Left-handed double helix with alternating purine-pyrimidine sequence
(B) Triple-stranded helix
(C) Parallel double helix
(D) Right-handed wide helix

6. Which chemical bond links adjacent nucleotides together in a DNA or RNA polynucleotide chain?

(A) Phosphodiester bond (3' to 5')
(B) Peptide bond
(C) Glycosidic bond
(D) Hydrogen bond

7. The classic Meselson-Stahl experiment (1958) demonstrated that DNA replication is:

(A) Semiconservative
(B) Conservative
(C) Dispersive
(D) Random

8. In E. coli, which enzyme is the primary replicative DNA polymerase responsible for both leading and lagging strand synthesis?

(A) DNA Polymerase III
(B) DNA Polymerase I
(C) DNA Polymerase II
(D) RNA Polymerase

9. Which enzymatic activity is responsible for the proofreading capacity of replicative DNA polymerases?

(A) 3' to 5' exonuclease activity
(B) 5' to 3' exonuclease activity
(C) Endonuclease activity
(D) DNA ligase activity

10. Which E. coli DNA polymerase possesses unique 5' to 3' exonuclease activity to remove RNA primers?

(A) DNA Polymerase I
(B) DNA Polymerase III
(C) DNA Polymerase II
(D) Klenow fragment

11. Short segments of DNA synthesized discontinuously on the lagging strand are known as:

(A) Okazaki fragments
(B) Kornberg fragments
(C) Watson fragments
(D) Crick fragments

12. The synthesis of short RNA primers required to initiate DNA replication is catalyzed by:

(A) Primase (DnaG in E. coli)
(B) RNA Polymerase II
(C) DNA Ligase
(D) Helicase

13. Which enzyme unwinds the double-stranded DNA duplex ahead of the replication fork?

(A) DNA Helicase (DnaB in E. coli)
(B) Topoisomerase
(C) Single-Strand Binding protein
(D) DNA Ligase

14. Single-stranded DNA generated at the replication fork is protected from nuclease degradation by:

(A) SSB in bacteria (RPA in eukaryotes)
(B) Histones
(C) PCNA
(D) Cohesin

15. Which sliding clamp protein encircles DNA to confer high processivity on eukaryotic DNA Polymerase delta and epsilon?

(A) PCNA (Proliferating Cell Nuclear Antigen)
(B) Beta clamp
(C) RFC
(D) MCM

16. The enzyme that seals nicks in the phosphodiester backbone between adjacent Okazaki fragments is:

(A) DNA Ligase
(B) DNA Polymerase I
(C) Topoisomerase I
(D) Exonuclease III

17. Which enzyme relieves torsional supercoiling stress generated ahead of the advancing replication fork?

(A) DNA Topoisomerase (Gyrase in bacteria)
(B) Helicase
(C) Primase
(D) Telomerase

18. Telomerase prevents chromosome shortening in germ and cancer cells by reverse-transcribing:

(A) Its own intrinsic RNA template (TERC)
(B) Cytoplasmic mRNA
(C) Ribosomal RNA
(D) Exogenous DNA

19. Xeroderma Pigmentosum is caused by inherited defects in which DNA repair pathway?

(A) Nucleotide Excision Repair (NER)
(B) Base Excision Repair (BER)
(C) Mismatch Repair (MMR)
(D) Homologous Recombination

20. The primary enzyme that initiates Base Excision Repair (BER) by recognizing damaged bases is:

(A) DNA Glycosylase
(B) AP Endonuclease
(C) DNA Helicase
(D) Polymerase beta

21. Lynch Syndrome (HNPCC) is an autosomal dominant cancer syndrome caused by germline mutations in:

(A) Mismatch Repair genes (MLH1, MSH2, MSH6, PMS2)
(B) BRCA1 and BRCA2
(C) TP53
(D) ATM

22. In prokaryotic transcription, promoter recognition by RNA polymerase core enzyme requires:

(A) Sigma factor
(B) Rho factor
(C) NusA protein
(D) TATA-binding protein

23. The conserved hexameric sequence located ~10 base pairs upstream of the transcription start site in bacteria is the:

(A) Pribnow box (-10 element: TATAAT)
(B) TATA box
(C) CAAT box
(D) GC box

24. Rho-independent (intrinsic) transcription termination in bacteria relies on the formation of a:

(A) GC-rich hairpin loop followed by a poly-U tract
(B) Poly-A tail
(C) Protein roadblock
(D) Pseudoknot

25. Which antibiotic specifically inhibits bacterial transcription by binding the beta-subunit of RNA polymerase?

(A) Rifampicin
(B) Streptomycin
(C) Chloramphenicol
(D) Tetracycline

26. Which eukaryotic RNA polymerase synthesizes messenger RNA (mRNA) and most snRNAs and microRNAs?

(A) RNA Polymerase II
(B) RNA Polymerase I
(C) RNA Polymerase III
(D) RNA Polymerase IV

27. Which eukaryotic RNA polymerase synthesizes 28S, 18S, and 5.8S ribosomal RNAs?

(A) RNA Polymerase I
(B) RNA Polymerase II
(C) RNA Polymerase III
(D) Mitochondrial RNA Pol

28. Transfer RNAs (tRNAs) and 5S ribosomal RNA are transcribed in eukaryotes by:

(A) RNA Polymerase III
(B) RNA Polymerase I
(C) RNA Polymerase II
(D) Telomerase

29. Alpha-Amanitin, a toxin from death cap mushrooms (Amanita phalloides), strongly inhibits:

(A) RNA Polymerase II at extremely low concentrations (nM)
(B) RNA Polymerase I
(C) Bacterial RNA Polymerase
(D) DNA Polymerase delta

30. The TATA-binding protein (TBP) is a universal component of which general transcription factor complex?

(A) TFIID
(B) TFIIB
(C) TFIIF
(D) TFIIH

31. Which general transcription factor possesses ATP-dependent DNA helicase and CTD kinase activities?

(A) TFIIH
(B) TFIID
(C) TFIIA
(D) TFIIE

32. The C-Terminal Domain (CTD) of the largest subunit of RNA Polymerase II consists of tandem repeats of the heptapeptide:

(A) YSPTSPS
(B) PKKKRKV
(C) RGD motif
(D) DEAD box

33. Phosphorylation of Serine-5 on the CTD heptapeptide repeats by CDK7 triggers:

(A) Promoter clearance and recruitment of the 5'-capping machinery
(B) Transcriptional termination
(C) Polyadenylation
(D) Splicing completion

34. Phosphorylation of Serine-2 on the CTD heptapeptide repeats by P-TEFb (CDK9) marks:

(A) Productive transcriptional elongation and recruitment of splicing/polyadenylation factors
(B) Transcription initiation
(C) Promoter binding
(D) Polymerase degradation

35. The 5'-cap added to nascent eukaryotic pre-mRNA consists of a:

(A) 7-Methylguanosine attached via an unusual 5'-to-5' triphosphate bridge
(B) Poly-A tail
(C) 3'-terminal cytosine
(D) Deoxythymidine cap

36. Polyadenylation of eukaryotic pre-mRNA involves cleavage and addition of 200-250 adenine residues catalyzed by:

(A) Poly(A) Polymerase (PAP)
(B) RNA Polymerase II
(C) DNA Ligase
(D) Terminal Transferase

37. The invariant dinucleotide sequences defining the 5' splice donor and 3' splice acceptor sites in pre-mRNA introns are:

(A) GU at 5' end and AG at 3' end
(B) AU and GC
(C) AG and GU
(D) CA and TG

38. The nucleophile that attacks the 5' splice site in the first transesterification step of splicing is the:

(A) 2'-OH of the invariant branch point Adenosine
(B) 3'-OH of guanosine
(C) 5'-phosphate of uracil
(D) Water molecule

39. The excised intron released during pre-mRNA splicing adopts which unique topological intermediate?

(A) Lariat structure with a 2'-5' phosphodiester branch
(B) Closed circular DNA
(C) Linear fragment
(D) Hairpin loop

40. The major spliceosome is composed of five small nuclear ribonucleoproteins (snRNPs):

(A) U1, U2, U4, U5, and U6
(B) U1, U2, U3, U7, U8
(C) U11, U12, U4atac, U6atac
(D) 18S, 28S, 5.8S, 5S

41. DNA Polymerase III holoenzyme in E. coli achieves processivity exceeding 100,000 nucleotides per binding event due to the:

(A) Beta sliding clamp dimer
(B) Tau subunit
(C) Alpha subunit
(D) Epsilon subunit

42. The clamp loader complex in E. coli (gamma complex) loads the beta clamp onto DNA by utilizing energy from:

(A) ATP binding to open the clamp and ATP hydrolysis to close it
(B) GTP hydrolysis
(C) Proton gradient
(D) DNA supercoiling

43. In eukaryotes, the leading strand is synthesized primarily by DNA Polymerase epsilon, while the lagging strand is synthesized by:

(A) DNA Polymerase delta
(B) DNA Polymerase alpha-primase
(C) DNA Polymerase beta
(D) DNA Polymerase gamma

44. DNA Polymerase alpha is unique among eukaryotic polymerases because it forms a complex with primase to synthesize:

(A) A chimeric primer of ~10 nt RNA followed by ~20 nt DNA
(B) Long genomic DNA
(C) Telomeres
(D) Mitochondrial DNA

45. The 'Polymerase Switch' during eukaryotic replication involves replacing Pol alpha with:

(A) Pol epsilon on leading strand and Pol delta on lagging strand via RFC clamp loading
(B) Pol beta
(C) Pol gamma
(D) Telomerase

46. Flap Endonuclease 1 (FEN1) processes Okazaki fragments in eukaryotes by cleaving:

(A) 5'-overhanging single-stranded RNA/DNA flaps generated when Pol delta displaces the primer
(B) Double-stranded DNA
(C) Telomeric loops
(D) Apurinic sites

47. Bacterial DNA Gyrase is a Type II topoisomerase that uniquely introduces:

(A) Negative supercoils into relaxed DNA utilizing ATP hydrolysis
(B) Positive supercoils
(C) Single-strand nicks only
(D) RNA primers

48. Quinolone antibiotics (e.g., Ciprofloxacin, Levofloxacin) kill bacteria by:

(A) Trapping the covalent Gyrase-DNA cleavable complex, causing double-strand breaks
(B) Inhibiting ribosome peptidyl transferase
(C) Blocking cell wall synthesis
(D) Inactivating RNA polymerase

49. Etoposide (VP-16) and Doxorubicin are human chemotherapeutics that target:

(A) Human Topoisomerase II, trapping the cleavable covalent complex
(B) Topoisomerase I
(C) DNA Ligase I
(D) Helicase

50. Camptothecin and its derivatives (Topotecan, Irinotecan) specifically inhibit:

(A) Human Topoisomerase I by stabilizing the covalent topoisomerase-DNA nicked intermediate
(B) Topoisomerase II
(C) DNA Polymerase alpha
(D) PCNA

51. In Base Excision Repair, cleavage of the N-glycosidic bond by uracil-DNA glycosylase leaves an:

(A) AP (Apurinic/Apyrimidinic) site
(B) Double-strand break
(C) Thymine dimer
(D) RNA flap

52. AP endonuclease-1 (APE1) cleaves the phosphodiester backbone at an AP site immediately:

(A) 5' to the abasic ribose, generating a 3'-OH and 5'-deoxyribose phosphate (5'-dRP)
(B) 3' to the site
(C) Across both strands
(D) In the major groove

53. In short-patch BER, DNA Polymerase beta fills the single-nucleotide gap and removes the 5'-dRP group using its:

(A) Intrinsic dRP lyase activity
(B) 3'-5' exonuclease
(C) Helicase activity
(D) Primase activity

54. Ultraviolet (UV) radiation damages DNA primarily by inducing covalent linkage between adjacent pyrimidines, forming:

(A) Cyclobutane Pyrimidine Dimers (CPDs) and 6-4 photoproducts
(B) Double-strand breaks directly
(C) 8-oxoguanine
(D) Abasic sites

55. In Global Genome Nucleotide Excision Repair (GG-NER), UV-induced CPD distortions are recognized by:

(A) XPC - RAD23B and the UV-DDB complex
(B) RNA Polymerase II
(C) Ku70/Ku80
(D) MutS-alpha

56. In Transcription-Coupled NER (TC-NER), lesions on the transcribed strand are detected by:

(A) Stalling of elongating RNA Polymerase II, which recruits CSB and CSA
(B) XPC
(C) UvrA
(D) DNA Polymerase delta

57. During eukaryotic NER, dual incisions excising a 24-30 nucleotide damaged oligonucleotide are executed by:

(A) XPG (cleaving 3') and the XPF-ERCC1 heterodimer (cleaving 5')
(B) AP endonuclease
(C) FEN1
(D) Caspase-3

58. Non-Homologous End Joining (NHEJ) initiates when the broken double-stranded DNA ends are captured by the ring-shaped heterodimer:

(A) Ku70 - Ku80
(B) BRCA1 - BRCA2
(C) Rad51
(D) Mre11 - Rad50

59. Ku70/80 recruits and activates which critical protein kinase during classical NHEJ?

(A) DNA-Dependent Protein Kinase catalytic subunit (DNA-PKcs)
(B) ATM kinase
(C) ATR kinase
(D) CDK1

60. In classical NHEJ, final ligation of trimmed double-stranded DNA ends is catalyzed by:

(A) DNA Ligase IV in complex with XRCC4 and XLF
(B) DNA Ligase I
(C) DNA Ligase III
(D) T4 DNA Ligase

61. Homologous Recombination repair of double-strand breaks initiates with 5'-to-3' end resection to generate:

(A) 3' single-stranded DNA overhangs coated with Rad51
(B) 5' single-stranded overhangs
(C) Blunt ends
(D) Hairpins

62. BRCA2 promotes homologous recombination by directly facilitating the loading of which recombinase onto ssDNA?

(A) Rad51
(B) Ku70
(C) DNA Polymerase beta
(D) XPG

63. Synthetic lethality in BRCA1/2-deficient cancers is achieved clinically using pharmacological inhibitors of:

(A) PARP1 (Poly(ADP-ribose) Polymerase 1)
(B) Topoisomerase II
(C) DNA Ligase IV
(D) ATM kinase

64. In the bacterial RNA polymerase holoenzyme (alpha2-beta-beta'-omega-sigma), the beta and beta' subunits form the:

(A) Catalytic claw that binds DNA and harbors the ribonucleotide active center
(B) Promoter recognition loop
(C) Dimerization domain
(D) Omega subunit

65. Sigma-70 recognizes the -35 element and -10 Pribnow element via which structural domains?

(A) Region 4.2 (recognizing -35) and Region 2.4 (recognizing -10)
(B) Region 1.1 and 1.2
(C) Region 3.1
(D) C-terminal domain

66. Sigma-32 (RpoH) directs bacterial RNA polymerase to transcribe specific promoters in response to:

(A) Heat shock stress
(B) Nitrogen starvation
(C) Iron deficiency
(D) Osmotic shock

67. Rho factor is a ring-shaped hexameric motor that terminates bacterial transcription by:

(A) Translocating 5'-to-3' along RNA and unwinding the RNA-DNA hybrid in an ATP-dependent manner
(B) Cleaving the RNA
(C) Phosphorylating RNA Pol
(D) Binding the promoter

68. TFIID consists of TBP (TATA-Binding Protein) and approximately 14 TAFs (TBP-Associated Factors) that recognize:

(A) Inr (Initiator), DPE (Downstream Promoter Element), and DCE motifs
(B) Upstream enhancers alone
(C) The poly-A signal
(D) Splice sites

69. TBP binds the minor groove of the TATA box and induces a sharp bend in DNA of approximately:

(A) 80 degrees
(B) 180 degrees
(C) 20 degrees
(D) 45 degrees

70. TFIIB bridges TFIID to RNA Polymerase II and precisely positions the catalytic cleft over the:

(A) Transcription start site (+1 nucleotide)
(B) TATA box
(C) -35 element
(D) Enhancer

71. The 5'-capping enzyme complex is recruited to the RNA Polymerase II CTD specifically when Ser-5 is phosphorylated, executing:

(A) RNA triphosphatase, guanylyltransferase, and (guanine-N7)-methyltransferase reactions
(B) Polyadenylation
(C) Exonuclease cleavage
(D) Splicing

72. The polyadenylation consensus signal recognized in nascent pre-mRNA by CPSF is:

(A) AAUAAA located 10-30 nucleotides upstream of the cleavage site
(B) UAG
(C) AUG
(D) TATAAA

73. Cleavage and polyadenylation specificity factor (CPSF) and cleavage stimulation factor (CstF) cleave pre-mRNA at a:

(A) CA dinucleotide site between the AAUAAA hexamer and a downstream GU-rich element
(B) Poly-U tract
(C) TATA site
(D) 5' cap

74. In the major spliceosome, U1 snRNP initiates spliceosome assembly by base-pairing with the:

(A) 5' splice donor site (conserved /GU sequence)
(B) 3' splice site
(C) Branch point adenosine
(D) Poly-A tail

75. U2 snRNP base-pairs with the branch point sequence, causing the branch point adenosine to:

(A) Bulge out of the duplex, exposing its 2'-OH for nucleophilic attack
(B) Form a triple helix
(C) Be methylated
(D) Dephosphorylate

76. The catalytic active site of all DNA and RNA polymerases contains two invariant Aspartate residues that coordinate:

(A) Two divalent magnesium ions (Mg2+ A and Mg2+ B) via a two-metal-ion catalytic mechanism
(B) Two zinc ions
(C) Two calcium ions
(D) Two iron atoms

77. In the two-metal-ion catalytic mechanism of DNA polymerases, Metal Ion A functions to:

(A) Lower the pKa of the 3'-OH of the primer, facilitating its deprotonation to an attacking nucleophile
(B) Coordinate incoming triphosphate
(C) Stabilize pyrophosphate
(D) Unwind DNA

78. In the two-metal-ion catalytic mechanism, Metal Ion B functions to:

(A) Coordinate the beta- and gamma-phosphates of dNTP, stabilizing the pentavalent transition state
(B) Deprotonate the 3'-OH
(C) Hydrolyze the primer
(D) Form a covalent bond

79. The kinetic fidelity of replicative DNA Polymerases (error rate ~10^-5 before proofreading) is primarily achieved through:

(A) Induced-fit geometric conformation of the active site fingers domain closing over correct Watson-Crick pairs
(B) Thermal diffusion
(C) Base stacking forces alone
(D) Direct reading of major groove

80. When a mismatched nucleotide is misincorporated, the stalling of polymerase translocation allows the mismatched 3' end to melt and partition into the:

(A) Exonuclease active site located ~30 A away from the polymerase active site
(B) Clamp loader
(C) PCNA ring
(D) Solvent directly

81. The human mitochondrial DNA polymerase (Pol gamma) consists of a catalytic subunit (Pol gamma-A) and an accessory dimer (Pol gamma-B) that:

(A) Increases processivity and DNA binding affinity by 100-fold
(B) Cleaves primers
(C) Synthesizes RNA
(D) Acts as a topoisomerase

82. Progressive External Ophthalmoplegia (PEO) and Alpers Syndrome are mitochondrial encephalomyopathies caused by mutations in:

(A) POLG (encoding DNA Polymerase gamma)
(B) POLA1
(C) POLD1
(D) POLE

83. Ultraviolet-induced photolesions that stall replicative DNA Polymerases are bypassed by low-fidelity Translesion Synthesis (TLS) polymerases:

(A) Pol eta, Pol iota, Pol kappa, and Rev1 (Y-family polymerases)
(B) Pol delta
(C) Pol epsilon
(D) Pol alpha

84. Xeroderma Pigmentosum Variant (XP-V) patients develop skin cancer without classical NER defects due to mutations in:

(A) POLH (encoding translesion DNA Polymerase eta)
(B) XPC
(C) XPA
(D) DDB2

85. The MCM2-7 replicative helicase forms a heterohexameric ring that unwinds DNA in which translocation polarity?

(A) 3' to 5' polarity along the leading strand template
(B) 5' to 3' polarity
(C) Bidirectional simultaneous
(D) Random polarity

86. In contrast to eukaryotic MCM2-7, the bacterial replicative helicase DnaB translocates along single-stranded DNA with:

(A) 5' to 3' polarity along the lagging strand template
(B) 3' to 5' polarity
(C) Zero polarity
(D) 3' to 3' polarity

87. The Shelterin complex protects telomeres from being recognized as double-strand breaks; TRF1 and TRF2 bind directly to:

(A) Double-stranded telomeric TTAGGG repeats via homeodomain-like Myb motifs
(B) Single-stranded overhang
(C) Centromeres
(D) RNA

88. POT1 (Protection of Telomeres 1) in the Shelterin complex binds specifically to the:

(A) Single-stranded 3' G-rich telomeric overhang
(B) Double-stranded T-loop
(C) Nuclear lamina
(D) Telomerase RNA

89. Dyskeratosis Congenita is an inherited bone marrow failure and premature aging syndrome caused by mutations in:

(A) TERT, TERC, or DKC1 (encoding dyskerin, which stabilizes telomerase RNA)
(B) BRCA1
(C) TP53
(D) DNA Ligase I

90. In human Mismatch Repair, MutS-alpha is a heterodimer composed of:

(A) MSH2 and MSH6 (recognizes single base-base mismatches and 1 bp insertion-deletion loops)
(B) MSH2 and MSH3
(C) MLH1 and PMS2
(D) MLH1 and MLH3

91. MutS-beta is a heterodimer of MSH2 and MSH3 specialized for recognizing:

(A) Larger insertion-deletion loops (2 to 8 nucleotides)
(B) Single mismatches
(C) Ribonucleotides
(D) AP sites

92. Microsatellite Instability (MSI-High) characterized by length alterations in short tandem repeats is the diagnostic hallmark of:

(A) Defective Mismatch Repair (dMMR)
(B) Defective NER
(C) Defective NHEJ
(D) Telomerase overexpression

93. During transcription initiation, bacterial RNA polymerase undergoes 'Abortive Initiation', repeatedly synthesizing and releasing:

(A) Short transcripts of 2 to 10 nucleotides before clearing the promoter
(B) Full-length mRNAs
(C) tRNAs
(D) Poly-A tails

94. Promoter clearance in bacteria occurs when the nascent RNA reaches ~10-12 nucleotides, which sterically displaces:

(A) Region 3.2 of the sigma factor from the RNA exit channel
(B) The beta subunit
(C) Magnesium
(D) DNA Gyrase

95. In eukaryotes, the transition of RNA Polymerase II from promoter-proximal pausing into productive elongation requires P-TEFb phosphorylation of:

(A) NELF (causing its dissociation) and DSIF (converting it into a positive elongation factor)
(B) TFIID
(C) TFIIA
(D) TBP

96. Flavopiridol (Alvocidib) is an experimental anticancer agent that arrests transcription elongation by specifically inhibiting:

(A) CDK9 (the catalytic kinase subunit of P-TEFb)
(B) CDK1
(C) CDK4
(D) Topoisomerase I

97. The catalytic core of the active spliceosome (B* and C complexes) is formed exclusively by an RNA network of:

(A) U2 and U6 snRNAs coordinating two catalytic magnesium ions, functioning as a Ribozyme
(B) Protein enzymes alone
(C) U1 and U4
(D) Histones

98. U4 snRNP acts as a negative chaperone that masks the catalytic domain of U6 snRNA until:

(A) The Brr2 RNA helicase unwinds the U4/U6 duplex, activating the catalytic center
(B) ATP is depleted
(C) The cap is added
(D) Lariat is cleaved

99. Spinal Muscular Atrophy (SMA) is an autosomal recessive motor neuron disease caused by deletion of SMN1; Nusinersen (Spinraza) treats SMA by:

(A) Acting as an antisense oligonucleotide that alters alternative splicing of SMN2 to include exon 7
(B) Inhibiting RNA Pol II
(C) Inactivating caspases
(D) Degrading SMN2

100. Cockayne Syndrome is a severe neurodevelopmental segmental progeroid disorder caused by mutations in ERCC6 (CSB) or ERCC8 (CSA), disrupting:

(A) Transcription-Coupled Nucleotide Excision Repair (TC-NER)
(B) Global Genome NER
(C) Base Excision Repair
(D) Telomerase

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Contrast the structural parameters of B-DNA, A-DNA, and Z-DNA regarding helical handedness, base pairs per turn, pitch, and major/minor groove topologies.

Q2. Detail the universal Two-Metal-Ion Catalytic Mechanism shared by all DNA and RNA polymerases.

Q3. Explain the mechanism of High-Fidelity DNA Replication, detailing the geometric induced-fit active site and 3' to 5' proofreading exonuclease.

Q4. Describe the molecular architecture of the eukaryotic Replication Fork, detailing the distinct roles of Pol alpha-primase, Pol delta, Pol epsilon, PCNA, RFC, and FEN1.

Q5. Explain the End-Replication Problem of linear chromosomes and detail how Telomerase preserves genomic stability.

Q6. Detail the Base Excision Repair (BER) pathway, distinguishing between Short-Patch and Long-Patch repair mechanisms.

Q7. Contrast Global Genome NER (GG-NER) and Transcription-Coupled NER (TC-NER) regarding damage recognition, molecular factors, and disease phenotypes.

Q8. Explain the two competing pathways for repairing Double-Strand DNA Breaks: Non-Homologous End Joining (NHEJ) versus Homologous Recombination (HR).

Q9. Describe the transcription initiation cycle in Prokaryotes, detailing the promoter recognition, open complex formation, and abortive initiation by RNA Polymerase.

Q10. Detail the stepwise assembly of the Eukaryotic Pre-Initiation Complex (PIC) for RNA Polymerase II, highlighting the general transcription factors TFIID, TFIIA, TFIIB, TFIIF, TFIIE, and TFIIH.

Q11. Explain the C-Terminal Domain (CTD) Phosphorylation Code of RNA Polymerase II and how it coordinates 5'-Capping, Splicing, and Polyadenylation.

Q12. Describe the two-step transesterification chemical mechanism of Pre-mRNA Splicing and the role of snRNAs in spliceosome catalysis.

Q13. Explain the molecular mechanism of Alternative Splicing regulation by SR proteins, hnRNPs, and exonic/intronic splicing enhancers (ESEs/ISEs).

Q14. Discuss the structural organization and catalytic cycle of DNA Topoisomerases, contrasting Type IA, Type IB, and Type IIA enzymes.

Q15. Explain the molecular architecture of the Shelterin complex and detail how it prevents telomeres from triggering the DNA damage response.

Q16. Detail the molecular mechanism of Transcription-Coupled Repair (TC-NER) and explain how Cockayne Syndrome proteins CSA and CSB operate.

Q17. Describe the Mismatch Repair (MMR) pathway in E. coli, highlighting the roles of MutS, MutL, MutH, and Dam methylase in strand discrimination.

Q18. Explain how human Mismatch Repair operates without hemimethylation, detailing MSH/MLH complexes and nick-directed exonucleolytic excision.

Q19. Discuss the molecular regulation of Transcription Pausing and Elongation by DSIF, NELF, and P-TEFb.

Q20. Detail the molecular mechanism of Rho-Dependent Transcription Termination in prokaryotes.

Q21. Explain the structural mechanics of the Spliceosome Catalytic Activation (B to B* and C complexes) driven by DEXD/H-Box RNA Helicases.

Q22. Describe the molecular pathology of Xeroderma Pigmentosum, Cockayne Syndrome, and Trichothiodystrophy as clinical manifestations of NER mutations.

Q23. Explain the chemical and physical basis of DNA Denaturation (Melting), Hyperchromicity, and the factors determining Melting Temperature (Tm).

Q24. Discuss the molecular mechanism of Homologous Recombination mediated by the RecBCD pathway and RecA in Escherichia coli.

Q25. Explain how the Mediator Complex integrates transcriptional activators with RNA Polymerase II and general transcription factors.

Q26. Describe the structural organization of Nucleosomes and explain how Histone Acetylation and Methylation regulate chromatin accessibility.

Q27. Explain the molecular mechanisms of Transcription Termination in Eukaryotes, contrasting the Allosteric (Conformational) and Torpedo models.

Q28. Describe the CRISPR-Cas9 bacterial adaptive immune system and detail how Cas9 introduces targeted double-strand breaks.

Q29. Explain the molecular mechanism of Telomere Protection by the T-Loop and describe how TRF2 suppresses classical Non-Homologous End Joining.

Q30. Describe the structure and mechanism of Group I and Group II Self-Splicing Introns, comparing their catalytic strategies to the nuclear Spliceosome.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

nucleophilic alkoxide ion (3'-O-).
3. Role of Metal Ion B (The Transition State Stabilizer):
- Metal Ion B is coordinated by the aspartates and coordinates all three phosphate groups (alpha, beta, and gamma) of the incoming dNTP.
- Function: It neutralizes the negative charges of the triphosphate moiety, correctly positions the alpha-phosphate for in-line nucleophilic attack, and stabilizes the pentacovalent transition state intermediate.
4. Chemical Execution and Pyrophosphate Release:
- The activated 3'-O- attacks the alpha-phosphorus atom of the incoming dNTP via a classic bimolecular nucleophilic substitution (SN2) reaction, inverting stereochemistry at phosphorus.
- A new 3'-5' phosphodiester bond is synthesized.
- Metal Ion B stabilizes the leaving group, inorganic Pyrophosphate (PPi).
- Pyrophosphatase immediately hydrolyzes PPi into two molecules of inorganic orthophosphate (2 Pi), providing a strong thermodynamic pulling force (Delta G0' ~ -19 kJ/mol) that drives the overall polymerization reaction irreversibly forward.

>>Solution to Question 3: Explain the mechanism of High-Fidelity DNA Replication, detailing the geometric induced-fit active site and 3' to 5' proofreading exonuc

>>Solution Question Describe the Eukaryotic Replication Problem of linear chromosomes and Telomerase. Pol delta, Pol e

two core components:
- TERT (Telomerase Reverse Transcriptase): The catalytic protein subunit possessing reverse transcriptase motifs.
- TERC (Telomerase RNA Component / hTR): An intrinsic, stable non-coding RNA containing a conserved 11-nucleotide template sequence (3'-CAAUCCCAAUC-5' in humans) complementary to the human telomeric hexamer repeat (5'-TTAGGG-3').
3. The Catalytic Elongation Cycle:
- Step 1 (Annealing): The single-stranded 3' G-rich DNA overhang of the chromosome binds and base-pairs with the 3'-portion of the TERC RNA template.
- Step 2 (Reverse Transcription): TERT uses its integral TERC template to synthesize six deoxyribonucleotides (dGTP, dTTP), extending the single-stranded 3'-end of the chromosome by one TTAGGG repeat.
- Step 3 (Translocation): TERT/TERC shifts forward along the newly synthesized DNA strand, re-aligning the template for another cycle of repeat synthesis.
- Step 4 (C-Strand Fill-In): Once the G-rich strand has been substantially extended, conventional Pol alpha-primase and Pol delta synthesize the complementary C-rich strand, followed by FEN1 processing and ligation.
4. Cancer and Cellular Immortality:
Normal human somatic cells repress TERT expression; telomeres progressively shorten, triggering the Hayflick limit and senescence. In contrast, > 90% of human malignancies upregulate TERT expression, acquiring unlimited replicative immortality.

Solution to Question 6:

strand displacement synthesis, polymerizing 2 to 10 nucleotides and peeling back a 5'-single-stranded DNA flap containing the blocking abasic lesion.

* Flap Endonuclease 1 (FEN1) binds PCNA and endonucleolytically cleaves the flap.
* DNA Ligase I seals the nick.

[Detailed Step-by-Step Answer]:
Double-Strand Break (DSB) Repair: NHEJ vs. Homologous Recombination:
1. Classical Non-Homologous End Joining (c-NHEJ):
- Cell Cycle Timing: Active throughout all cell cycle phases (G0, G1, S, G2); dominant in G0/G1 where sister chromatids are absent.
- Fidelity: Error-prone; direct ligation of ends frequently involves nucleolytic trimming and template-independent insertions, introducing small deletions and point mutations.
- Molecular Cascade:
* Step 1: The doughnut-shaped Ku70 - Ku80 heterodimer binds with picomolar affinity to the exposed blunt or jagged double-stranded DNA ends.
* Step 2: Ku70/80 recruits the DNA-Dependent Protein Kinase catalytic subunit (DNA-PKcs), assembling the active DNA-PK holoenzyme that synapsed the two DNA ends.
* Step 3: Complex end-processing: Artemis endonuclease is activated by DNA-PKcs phosphorylation to trim non-ligatable hairpins and single-strand overhangs.
* Step 4: The specialized ligase complex DNA Ligase IV - XRCC4 - XLF joins the broken strands.
2. Homologous Recombination (HR):
- Cell Cycle Timing: Strictly restricted to S and G2 phases when an identical, undamaged sister chromatid is available as a repair template.
- Fidelity: High-fidelity, error-free repair.
- Molecular Cascade:
* Step 1 (End Resection): The MRN complex (Mre11-Rad50-Nbs1) and CtIP initiate nucleolytic resection, expanded by Exo1 and Dna2 helicase, digesting 5'-strands to expose long 3' single-stranded DNA overhangs (> 1 kb).
* Step 2 (Rad51 Nucleofilament Assembly): The ssDNA is coated with RPA. BRCA2 displaces RPA, catalyzing the cooperative assembly of the recombinase Rad51 into a right-handed helical nucleoprotein filament.
* Step 3 (Strand Invasion and D-Loop): The Rad51-ssDNA filament scans the sister chromatid for homologous sequence, invading the duplex to form a displacement loop (D-loop).
* Step 4 (Synthesis and Resolution): DNA Polymerase delta extends the invading 3'-end using the sister chromatid as template. The structure is resolved either by Synthesis-Dependent Strand Annealing (SDSA, non-crossover) or by Double Holliday Junction resolution (via BTR or Gen1/Mus81 resolvases).

>> **Solution to Question 9:** Describe the transcription initiation cycle in Prokaryotes, detailing the promoter recognition, open complex formation, and abortive initia

3. Transition to the Open Complex (RPo / DNA Melting):
- Without requiring ATP hydrolysis, the holoenzyme undergoes a spontaneous thermodynamic isomerisation driven by protein conformational change.
- Invariant aromatic residues in Region 2.3 of sigma-70 (Tyr-430 and Phe-427) flip the base Adenosine at position -11 and Adenosine at -7 out of the DNA duplex into hydrophobic pockets of the protein.
- This base flipping nucleates localized melting of ~14 base pairs between positions -11 and +3, exposing the single-stranded template strand in the catalytic active site cleft, forming the transcriptionally competent Open Complex (RPo).
4. Abortive Initiation and Promoter Clearance:
- In the open complex, RNA polymerase binds the initial initiating ribonucleoside triphosphate (typically ATP or GTP at +1) and synthesizes short transcripts of 2 to 9 nucleotides.
- However, during early synthesis, the growing transcript collides sterically with the N-terminal 'sigma-finger' loop (Region 3.2), which obstructs the nascent RNA exit channel.
- The polymerase repeatedly releases these short abortive RNA fragments (2-9 nt), remaining stationary at the promoter while reeling in downstream DNA ('DNA scrunching').
- When the nascent transcript successfully reaches ~10-12 nucleotides, its growing bulk mechanically pushes the sigma-finger out of the exit channel.
- Contact between sigma and the core enzyme is disrupted, sigma factor dissociates, and the core enzyme 'clears' the promoter, transitioning into a highly stable, processive Ternary Elongation Complex (TEC).

>>**Solution to Question 18:** the stepwise assembly of the Eukaryotic Pre-Initiation Complex (PIC) for RNA Polymerase II, highlighting the general transcriptio

- The CDK7 kinase subunit of the CAK module phosphorylates Serine-5 of the Pol II C-Terminal Domain (CTD).
- Ser-5 phosphorylation disrupts electrostatic contacts between Pol II and general transcription factors (TFIID, TFIIB), triggering Promoter Clearance and recruiting the 5'-capping enzyme complex to initiate productive transcription.

>>Solution to Question 12: Describe the C-Terminal Domain (CTD) of RNA Pol II and the role of its Serine-5 phosphorylation in transcription initiation.

- 5' Splice Donor Site: Invariant 5'-GU-3' dinucleotide embedded in consensus /GURAGU.
- 3' Splice Acceptor Site: Invariant 5'-AG-3' dinucleotide preceded by a Polypyrimidine Tract (PPT, ~15-20 consecutive U/C residues).
- Branch Point Sequence: Conserved sequence located 20-50 bp upstream of the 3' site containing an Invariant Adenosine (YNYURAY).

2. The Two Sequential Transesterification Steps:
Nuclear pre-mRNA splicing is executed via two consecutive S_N2-type transesterification reactions that conserve the total number of phosphodiester bonds (requiring zero high-energy ATP bond cleavage for the chemical ligation steps):
- Step 1 (First Transesterification - Lariat Formation):
* The 2'-hydroxyl group (2'-OH) of the invariant Branch Point Adenosine acts as a nucleophile, attacking the phosphodiester bond at the 5' splice site (the G of the GU intron boundary).
* Cleavage occurs at the 5' exon-intron junction, freeing Exon 1 with a terminal 3'-OH group.
* Concurrently, a novel, unusual 2'-5' phosphodiester bond is formed between the 2'-OH of the branch point adenosine and the 5'-phosphate of the first intron nucleotide (G), generating the circularized 'Lariat' intermediate.
- Step 2 (Second Transesterification - Exon Ligation):
* The newly liberated 3'-OH of Exon 1 acts as a nucleophile, attacking the phosphodiester bond at the 3' splice site (immediately following the AG dinucleotide).
* The two exons are covalently joined via a standard 3'-5' phosphodiester bond.
* The excised intron is released as a free Lariat RNA, which is rapidly linearized by the debranching enzyme (DBR1) and degraded by nuclear ribonucleases.

3. The Spliceosome as a Ribozyme:
High-resolution cryo-EM structures demonstrate that the spliceosome is an authentic RNA-based catalytic enzyme (Ribozyme):
- U2 and U6 snRNAs base-pair into an intricate secondary structure coordinating two catalytic divalent Magnesium ions (Mg2+ A and Mg2+ B).
- The U2/U6 RNA network perfectly mirrors the active-site catalytic domain V of self-splicing Group II introns, proving that eukaryotic nuclear splicing is catalytically homologous to ancient self-splicing ribozymes.

>>Solution to Question 13 Explain the molecular mechanism of Alternative Splicing regulation by SR proteins, hnRNPs, and exonic/intronic splicing enhancers (ES

arginine and serine dipeptides.
* Activator Mechanism: SR proteins bound to ESEs utilize their RS domains to recruit and stabilize U1 snRNP at the weak 5' splice site and recruit U2AF (U2-Auxiliary Factor) to the 3' polypyrimidine tract/AG site, promoting exon inclusion.
- Heterogeneous Nuclear Ribonucleoproteins (hnRNPs, e.g., hnRNP A1, hnRNP I / PTB):
* Structural Organization: Possess multiple RRMs or KH domains but lack RS domains.
* Repressor Mechanism: Bind sequence-specifically to ESS and ISS silencer motifs. Bound hnRNPs oligomerize cooperatively along the pre-mRNA, looping out exons or sterically masking splice sites, preventing U1 and U2 snRNP docking, thereby driving Exon Skipping.
4. Clinical Example (SMN1/SMN2 and Spinal Muscular Atrophy):
- Humans have two near-identical genes: SMN1 (telomeric) and SMN2 (centromeric).
- A single silent C-to-T transition in exon 7 of SMN2 (c.840C>T) disrupts an SF2/ASF ESE while creating an hnRNP A1 ESS silencer.
- This causes 90% of SMN2 transcripts to skip exon 7, producing a truncated, unstable SMNDelta7 protein.
- Deletion of SMN1 leads to Spinal Muscular Atrophy (SMA), the leading genetic cause of infant death.
- The therapeutic antisense oligonucleotide Nusinersen (Spinraza) binds to an ISS silencer in intron 7 of SMN2, physically displacing hnRNP A1, restoring exon 7 inclusion and producing therapeutic levels of full-length SMN protein.

>>Solution to Question 14: Discuss the structural organization and catalytic cycle of DNA Topoisomerases, contrasting Type IA, Type IB, and Type IIA enzymes.

linkages, spaced 4 base pairs apart.
- Mechanism: ATP-Dependent Two-Gate Strand Passage. Binding of two ATP molecules to the N-terminal ATPase domains closes an N-gate, trapping an intact second DNA duplex (the T-segment / transport segment). ATP hydrolysis drives the T-segment through the open G-segment break and out a C-terminal exit gate, followed by G-segment religation.
- Changes linking number in discrete steps of Delta Lk = +/- 2 per cycle.

- When elongating RNA Polymerase II encounters a bulky DNA photoproduct (e.g., UV-induced CPD) on the template strand, the lesion cannot enter the catalytic active site cleft.
- Pol II stalls irreversibly, occluding ~35 base pairs of DNA around the lesion, paradoxically preventing the standard GG-NER sensor (XPC) from gaining access.

3. Recruitment of Cockayne Syndrome Proteins (CSB and CSA):
- Stalled Pol II is recognized by Cockayne Syndrome Protein B (CSB / ERCC6):
CSB is an SWI/SNF-like ATP-dependent chromatin remodeler that binds tightly to the upstream face of stalled Pol II and wraps adjacent DNA.
- CSB recruits Cockayne Syndrome Protein A (CSA / ERCC8):
CSA is a WD40-repeat protein functioning as part of a cullin-RING E3 ubiquitin ligase complex (CRL4-CSA).

4. Backtracking and Core Machinery Recruitment:
- CSB and CSA recruit the histone acetyltransferase p300 and the multi-protein complex UVSSA (UV-Stimulated Scaffold Protein A).
- UVSSA recruits the deubiquitinating enzyme USP7, which stabilizes CSB against proteasomal degradation.
- Working cooperatively, these factors remodel stalled Pol II, inducing reverse translocation (backtracking) by ~10 base pairs, or facilitating Pol II ubiquitination and removal, thereby exposing the buried DNA lesion.
- TFIIH is directly recruited to the site independently of XPC.
- XPA, RPA, XPG, and XPF-ERCC1 execute the standard dual incisions, excising the damaged 24-30 nt oligonucleotide, followed by gap repair synthesis by Pol delta/epsilon and ligation.

>>Solution to Question 17: Describe the Mismatch Repair (MMR) pathway in E. coli, highlighting the roles of MutS, MutL, MutH, and Dam methylase in strand disc

- Exonucleases degrade the displaced single strand:
* If MutH nicked 5' to the mismatch: RecJ (5'-to-3' exonuclease) or Exonuclease VII digests the strand.
* If MutH nicked 3' to the mismatch: Exonuclease I (3'-to-5') degrades it.
- Single-strand DNA is protected by SSB.
- DNA Polymerase III holoenzyme resynthesizes the excised patch, and DNA Ligase seals the nick.

>>Solution to Question 18: Mismatch Repair in E. coli...

- As the nascent transcript reaches ~20-30 nucleotides and receives its 5'-cap, two negative elongation factor complexes bind Pol II:
* DSIF (DRB Sensitivity-Inducing Factor): A heterodimer of Spt4 and Spt5.
* NELF (Negative Elongation Factor): A four-subunit complex (NELF-A, B, C/D, E).
- NELF-E binds the nascent RNA, while DSIF binds across the DNA-binding cleft of Pol II, mechanically locking the catalytic jaw and preventing forward translocation, freezing Pol II in a paused, unelongating state.
3. Physiological Purpose:
Maintains promoters in an open, nucleosome-free configuration, primed for instantaneous, synchronous, genome-wide transcriptional activation in response to heat shock, hormone pulses, or developmental signals.
4. The P-TEFb Release Switch:
- Upon receipt of activating signals, the positive elongation factor P-TEFb (Positive Transcription Elongation Factor b, composed of Cyclin T1 and CDK9) is recruited to paused promoters by transcription factors (e.g., c-Myc, NF-kappa-B) or the Super Elongation Complex (SEC).
- Active CDK9 executes three coordinated phosphorylation events:
* Phosphorylates the NELF-E subunit: Phosphorylation introduces negative charges, causing NELF to dissociate completely from Pol II into the nucleoplasm.
* Phosphorylates the C-terminal repeat domain of Spt5 (DSIF): Phosphorylation converts DSIF from a negative repressor into a powerful POSITIVE elongation factor that travels with Pol II.
* Phosphorylates Serine-2 of the RNA Pol II CTD: Shifts Pol II into productive, high-velocity elongation (> 2-3 kb/minute) and licenses co-transcriptional splicing.

Solution to Question 28: Explain the molecular mechanism of Rho-Dependent Transcription Termination in prokaryotes.

- Downstream of the rut site, the template contains specific pause sequences that cause RNA polymerase to temporarily pause.
- Rho catches up to the paused RNA polymerase complex.
- Its robust ATP-dependent RNA-DNA helicase activity mechanically pries open the 8-9 base pair RNA-DNA hybrid in the catalytic center, stripping the nascent transcript out of the polymerase cleft and causing core enzyme dissociation.

(CPDs) across non-transcribed genomic DNA. Unrepaired lesions persist through S-phase, forcing low-fidelity error-prone translesion polymerases to bypass them, resulting in massive C-to-T transition mutation burdens (the UV mutational signature).
- Clinical Features: Extreme cutaneous photosensitivity, poikiloderma, severe solar lentigines (freckling), and a > 10,000-fold elevated incidence of basal cell carcinomas, squamous cell carcinomas, and lethal cutaneous melanomas before age 10.
2. Cockayne Syndrome (CS - The Transcription-Deficiency Syndrome):
- Genetic Etiology: Mutations in ERCC6 (CSB) or ERCC8 (CSA), strictly disrupting Transcription-Coupled NER.
- Molecular Pathogenesis: While GG-NER is intact (allowing clearance of most DNA lesions), stalled RNA Polymerase II cannot be cleared from the template strand of transcribed genes. This triggers chronic, widespread transcriptional arrest, activating p53-dependent apoptotic death in post-mitotic tissues.
- Clinical Features: NO increased skin cancer risk! Patients present with progressive, severe microcephaly, developmental dwarfism, bird-like sunken eyes, sensorineural deafness, retinal pigmentary degeneration, and catastrophic demyelinating leukodystrophy resulting in death in early childhood.
3. Trichothiodystrophy (TTD - The Basal Transcription Defect):
- Genetic Etiology: Specific point mutations in the XPB, XPD, or p44 subunits of the General Transcription Factor TFIIH.
- Molecular Pathogenesis: The mutations do not completely destroy TFIIH helicase activity; instead, they destabilize the architectural framework of the TFIIH complex. This impairs both NER and the basal transcription of highly expressed structural genes.
- Clinical Features: Characterized by Sulfur-Deficient Brittle Hair and Nails (due to impaired transcription of cysteine-rich keratin genes, showing a diagnostic alternating 'tiger-tail' banding pattern under polarizing microscopy), congenital ichthyosis, intellectual disability, and photosensitivity, with zero elevation in skin cancer risk.

>> **Solution to Question 23:** Explain the chemical and physical basis of DNA Denaturation (Melting), Hyperchromicity, and the factors determining Melting Tempera

Higher %GC linearly elevates Tm (Marmur-Doty equation: Tm = 69.3 + 0.41 * [%GC]).
- Monovalent Cation Concentration (Ionic Strength, [Na+]): The phosphate backbone carries a dense negative charge (-1 per nucleotide). In pure water, mutual electrostatic repulsion between opposing strands destabilizes the duplex. Increasing salt concentration shields negative charges, dramatically stabilizing the duplex and raising Tm (Delta Tm proportional to log10[Na+]).
- Duplex Length: Longer duplexes possess more total base-stacking interactions, raising Tm up to a plateau (~500 bp).
- Denaturing Solvents: Formamide, urea, and dimethyl sulfoxide (DMSO) form competing hydrogen bonds with bases, linearly depressing Tm.

The Mediator Multi-Subunit Coactivator Complex:

1. Structural Architecture of Mediator:

Mediator is a massive (~1.4 MDa), conserved multiprotein coactivator complex consisting of 26 to 30 distinct subunits organized into four modular structural domains:

- Head Module: Directly contacts the unphosphorylated C-terminal domain (CTD) of RNA Polymerase II, TBP, and TFIIB.
- Middle Module: Forms an extensive structural bridge contacting Pol II subunits Rpb4/Rpb7 and TFIIE.
- Tail Module: Consists of the most evolutionarily divergent subunits (Med15, Med16, Med23), serving as the physical sensor that directly binds diverse sequence-specific transcription factor activation domains.
- Transient CDK8 Kinase Module (CDK8, Cyclin C, Med12, Med13): Reversibly associates with Mediator, functioning predominantly as a transcriptional repressor by sterically occluding Pol II interaction and phosphorylating Cyclin H of TFIIH.

2. Molecular Function as a Molecular Conduit:

- Sequence-specific transcription factors (activators) bound to distal enhancers (> 50 kb away) loop across chromatin to contact specific subunits in the Mediator Tail (e.g., SREBP binds Med15; nuclear hormone receptors bind Med1).
- This binding induces cooperative conformational rearrangements that propagate from the Tail through the Middle module to the Head module.
- The rearranged Head module binds with high affinity to RNA Polymerase II, stabilizing its assembly into the Pre-Initiation Complex (PIC) and stimulating TFIIH-dependent open complex formation.

3. Transcriptional Re-Initiation Scaffolding:

Upon promoter clearance, Pol II escapes into elongation, but the Mediator complex remains docked at the promoter alongside TFIID, forming a stable 'Scaffold' that recruits subsequent rounds of RNA Polymerase II molecules, driving rapid transcriptional re-initiation.

>> **Solution to Question 26:** Describe the structural organization of Nucleosomes and explain how Histone Acetylation and Methylation regulate chromatin accessib

- Effector Recruitment: Acetylated lysines serve as specific docking platforms recognized by Bromodomain-containing proteins, including SWI/SNF chromatin remodelers and TFIID, driving active transcription.

3. Histone Methylation (Context-Dependent Signaling):

- Catalyzed by Histone Methyltransferases (HMTs); reversed by Histone Demethylases (KDMs).
- Unlike acetylation, methylation does NOT alter the electrical charge of the lysine residue! Instead, it adds 1, 2, or 3 methyl groups (mono-, di-, trimethylation), altering steric bulk and hydrophobicity.
- Context-Dependent Transcriptional Meaning:

* H3K4me3: Localized at active gene promoters; recognized by Chromodomain and PHD-finger proteins that recruit the transcription initiation machinery.
* H3K9me3: Found in constitutive heterochromatin; recruits Heterochromatin Protein 1 (HP1) to enforce permanent gene silencing.
* H3K27me3: Polycomb-mediated facultative heterochromatin mark; recruits Polycomb Repressive Complex 1 (PRC1) to silence developmental loci.

>>Solution to Question 27: Explain the molecular mechanisms of Transcription Termination in Eukaryotes, contrasting the Allosteric (Conformational) and Torpedo

ample time to catch up and strip the transcript, executing efficient termination.

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double-stranded DNA helix. Because the free end is topological trapped inside a D-loop, it cannot be recognized as a substrate by the Ku70/80 ring or the MRN double-strand break sensor.

3. TRF2 as the Master Inhibitor of Classical NHEJ:

- TRF2 binds with high affinity to double-stranded TTAGGG repeats via its C-terminal Myb domain.
- Dimerization of TRF2 bends telomeric DNA by ~120 degrees, promoting the looping of the telomere back onto itself to facilitate T-loop formation.
- Direct Ku70/80 Blockade: TRF2 physically blocks the Ku70/80 heterodimer from encircling the chromosome terminus.
- ATM Kinase Inactivation: TRF2 possesses an N-terminal basic domain that directly binds the checkpoint sensor kinase ATM, locking it in an inactive, unphosphorylated state.

4. Pathological Consequence of TRF2 Deletion:

Experimental conditional knockout or dominant-negative inhibition of TRF2 causes instantaneous, catastrophic T-loop unravelling:

- Chromosome ends are instantly recognized as authentic double-strand breaks.
- Ku70/80 loads immediately onto every telomere end, recruiting DNA-PKcs and DNA Ligase IV.
- Massive end-to-end chromosome fusions occur, ligating telomeres together to form giant multicentric ring and dicentric chromosomes.
- In subsequent mitoses, dicentric chromosomes form anaphase bridges that shatter (Breakage-Fusion-Bridge cycles), driving massive genomic instability.

>>Solution to Question 30: Describe the structure and mechanism of Group I and Group II Self-Splicing Introns, comparing their catalytic strategies to the nuclear S

4. Structural Homology to the Nuclear Spliceosome:
- Group II introns and the eukaryotic spliceosome are universally acknowledged to share direct common ancestry:
* Both proceed via identical two-step transesterification chemistry using a 2'-OH branch point adenosine nucleophile to form a lariat intermediate.
* The catalytic core of Domain V in Group II introns is structurally superimposable on the U2/U6 snRNA catalytic duplex in the activated spliceosome.
* Over evolutionary time, the structural domains of ancient Group II introns fragmented in trans, evolving into the independent U1, U2, U4, U5, and U6 snRNAs, while the host cell supplied stabilizing proteins to form the modern spliceosome.
